MIHIR KHATRI

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SUMMARY

Third-year CS undergraduate (GPA 9.4/10) working across LLM systems, agentic AI, and GPU-aware inference. First-author preprint, two ongoing research internships, and several shipped systems. Careful about evaluation (large multi-subset benchmarks, significance tests, ablations) and about turning research code into working products.

EDUCATION

Sardar Patel Institute of Technology, Mumbai University | *B.Tech, Computer Science and Engineering* Expected Aug 2028
GPA 9.4/10.0. Algorithms, Theory of Computation, Operating Systems, Computer Networks, Probability & Statistics, Linear Algebra.

RESEARCH

Ro-SVD: Position-Aware SVD Compression of KV Caches in GQA Models | *first-author preprint, Zenodo* May 2026

- Proposed online low-rank compression of pre-RoPE key tensors in grouped-query attention, with adaptive per-head rank allocation and a custom CUDA de-rotation kernel plus randomised SVD path ($O(Thd^2)$ to $O(Thr \log r)$).
- **73.91% ROUGE-L win rate** vs. the uncompressed baseline at **35.1% less KV-cache memory**, over 1,602 samples on all 16 LongBench subsets (Qwen2.5-7B-Instruct, 4-bit NF4, dual T4).

IIITDM Kancheepuram: Research Intern | *Prof. Rajesh Kolluri* Sep 2026 – Present

- Designed the hypothesis and experiments for a differentiable residual-Dyna world model for safe vehicle-to-grid control: analytical AC power-flow base plus learned residual, QP hard-safety projection layer, FedProx personalisation.

IIT Jodhpur: Research Intern | *Prof. Deeksha Varshney* Sep 2026 – Present

- Equal contributor on hypothesis design, experiments, and evaluation of distilled small language models on multilingual tasks, extending Microsoft's Phi family.

INDUSTRY EXPERIENCE

Daten & Wissen Pvt. Ltd.: AI/ML Intern | *Thane, India* 2025

- Fine-tuned PaddleOCR for license plate detection and recognition (~88% end-to-end accuracy on 1,000 test images); improved robustness on degraded inputs; benchmarked U-Net, SAM, and ResNet for latency vs. accuracy; ran Optuna sweeps.

PROJECTS

INDRA + Summertime-server: Air-Gapped AI Workbench | *team project* SIH 2026

- Offline-first inference gateway (FastAPI, OpenAI-compatible API, RBAC) routing across vLLM, ONNX Runtime, ComfyUI, and llama-swap, with a **GPU VRAM lifecycle manager** (LRU eviction, queueing, concurrency caps) and a **sandboxed code executor** (Docker + gVisor, no network).
- Analyst client with a tool-calling agent orchestrator, React dashboard, CLI, offline voice, and local DOCX/PDF/PPTX/XLSX generation. Code: Indra_lok, Summertime-server.

Sentinel: Multi-Agent LLM Red-Teaming Harness | *LangGraph, OpenRouter, NVIDIA NIM* 2026

- Probes LLMs across five attack categories, with a cross-modal "modality gap" extension (text, image, video); results compared with McNemar and Wilcoxon tests.

ZENSO: Neural Engagement Analytics Platform | *LangGraph, Meta FAIR TRIBE v2* 2026 (ongoing)

- Lead ML developer on a platform that models viewer engagement from video. Six-module multi-agent architecture; ablation showed 16 frames keep **>0.99 Pearson r** at a **7.5x** inference speedup.

SkillFlow AI | CivicFlow | FAR AWAY 2026 form-filling agent | *LangGraph, Playwright, Selenium, Claude Vision* 2025–2026

- **SkillFlow AI**: deployed agentic skill-bartering platform for the JPMC Agastya Foundation, matching learners and mentors with zero manual moderation.
- **CivicFlow / FAR AWAY**: agents that find a target form, infer its fields, and generate browser automation to fill it from a one-time user profile.
- **Setu**: offline Bluetooth-mesh disaster-response messaging app with SOS relay, extending bitchat (demoable APK). **Mineguard**: real-time rockfall detection with YOLO. **Transformer from scratch** in PyTorch.

COMPETITIONS, OPEN SOURCE, OTHER

- **Smart India Hackathon 2026**: qualified SPIT's internal rounds; national-level result awaited.
- **Amazon ML Challenge 2026**: 99.999 accuracy on validation with a concurrent multi-LLM pipeline for entity record resolution, in place of a conventional model.
- **Kaggle IMAGINE (MEG decoding)**: ranked 47/176 with an Elastic Net decoder, SMOTE, and session domain adaptation.
- **Open source:** ML4SCI/DeepLense-AI-Scientist: 2 open PRs (GPU device auto-selection, checkpoint-metrics fix). **Problem solving:** 100+ LeetCode problems; FIDE 1700; SPIT chess champion. **Volunteer** mentor, JPMC Agastya Foundation.

SKILLS

Languages: Python, C/C++, Java, SQL, TypeScript, CUDA. **ML and systems:** PyTorch, TensorFlow, Hugging Face, LoRA/PEFT, 4-bit quantisation, Triton, cuBLAS, vLLM, FastAPI, Docker, React. **Agents:** LangGraph, tool-calling, multi-agent orchestration, Playwright/Selenium; AI coding tools (Claude, Codex, Cursor, Antigravity, Lovable, Bolt). **Evaluation:** LongBench, significance testing, ablations, Optuna. **Certificates:** Hugging Face Agents and NLP, NVIDIA DLI Agentic AI, Google Cloud ML APIs.